

RedBus

Online Bus Ticket Booking System

Project Documentation Report

Submitted By	Pasupuleti Balasai
Roll No	AV.SC.U4CSE25224
Secton	CSE-C
Department	Computer Science & Engineering
Year	1 st Year

1. Abstract

The redBus Clone is a front-end web application that replicates the core features of the popular bus-ticketing platform redBus.in. Built using HTML5, CSS3, and Vanilla JavaScript — with no backend or framework — it allows users to search bus routes, browse results, select seats from an interactive layout, and simulate booking. A login/sign-up modal with client-side validation is also included. This project is submitted as part of the Web Development course requirements.

2. Introduction

Online bus booking platforms like redBus have made travel planning simple and convenient. This project replicates such a platform on the front end to demonstrate how complex, data-driven web interfaces can be built using only native browser technologies. The application is a Single-Page Application (SPA) where all views coexist in one HTML file and are switched dynamically using JavaScript — no page reload is needed. All bus data and user state are stored in memory using JavaScript arrays and objects.

3. Objectives

- Build a multi-page SPA using only HTML, CSS, and JavaScript.
- Implement city autocomplete search with real-time filtering.
- Display dynamic bus results with filter options.
- Create an interactive seat map with colour-coded seat states.
- Calculate and display real-time fare summary.
- Develop a login/sign-up modal with client-side form validation.
- Make the application fully responsive for mobile and desktop.

4. Technologies Used

- HTML5 — Page structure and semantic elements.
- CSS3 — Styling, Flexbox/Grid layouts, animations, and CSS custom properties (variables) for theming.
- Vanilla JavaScript (ES6+) — SPA routing, DOM manipulation, event handling, and data rendering.
- Google Fonts — DM Sans (body) and DM Serif Display (headings).

5. Modules Description

Module 1 — Search

Landing page with city autocomplete, date picker, and passenger selector. A swap button exchanges the source and destination.

Module 2 — Results

Displays bus cards filtered from the in-memory dataset. Filter pills allow narrowing by AC, Non-AC, Sleeper, Seater, or Night departure.

Module 3 — Seat Selection

Interactive lower and upper deck seat map. Seats are colour-coded (available, booked, ladies, selected). Real-time fare summary updates on each selection.

Module 4 — Offers

Scrollable carousel of promotional offer cards with filter pills and navigation arrows on the home page.

Module 5 — Bookings

Booking history page with status filters (All, Upcoming, Completed, Cancelled). Shows an empty-state message when no bookings exist.

Module 6 — Authentication

Modal-based Login and Sign Up with fields for name, email, mobile number, and password. Validates input on the client side and updates the header to show user info on success.

6. Features Implemented

- City autocomplete with 30+ Indian cities
- Bus search with dynamic results rendering
- Bus filter pills (AC, Non-AC, Sleeper, Seater, Night)
- Interactive two-deck seat map (Lower + Upper)
- Real-time fare calculation with service fee
- Boarding and dropping point selectors
- Offers carousel with navigation dots
- Login / Sign Up modal with inline validation
- Password show/hide toggle
- Authenticated header with user avatar and dropdown
- Responsive layout for mobile and desktop
- SPA navigation (no page reload)

7. Workflow

1. User opens the app on the Search page. 2. Enters source/destination cities and travel date. 3. Clicks Search Buses — results page loads with matching buses. 4. Applies filters if needed; clicks View Seats on a bus. 5. Selects seats from the interactive map; fare summary updates live. 6. Chooses boarding and dropping points; clicks Proceed to Payment — confirmation shown. 7. User can also open the Login/Sign Up modal at any time to authenticate.

8. Requirements

Functional Requirements

- Search buses by source city, destination city, and date.
- Display and filter bus results dynamically.
- Allow seat selection with a maximum of 6 seats.
- Validate login/sign-up form fields with clear error messages.

9. Conclusion

The redBus Clone successfully demonstrates how a fully functional, feature-rich bus-booking web application can be built using only HTML, CSS, and JavaScript. It covers key front-end concepts — SPA navigation, dynamic data rendering, interactive UI components, form validation, and responsive design — making it a comprehensive and practical project. With further development incorporating a backend, real authentication, and payment integration, it could evolve into a production-ready applicatio

10. References

- redBus.in — <https://www.redbus.in>